UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

PART A (10 MARKS)

Answer ALL questions.

1. Which statement is most suitable for selecting among several constant cases?
   A. for
   B. do...while
   C. while
   D. switch

2. Which operator reads data from cin?
   A. %
   B. =
   C. >>
   D. <<

3. Which values can a bool variable hold?
   A. true or false
   B. any decimal
   C. any character
   D. any integer

4. How many times does a while loop execute when its condition is false initially?
   A. Until the user stops it
   B. Two
   C. One
   D. Zero

5. What is pseudocode?
   A. A structured English-like description of an algorithm
   B. A type of compiler error
   C. Machine code produced by an assembler
   D. A C++ header file

6. Which phase of the program development life cycle defines what the program must do?
   A. Coding
   B. Maintenance
   C. Analysis
   D. Testing

7. Which keyword prevents a variable's value from being changed after initialisation?
   A. break
   B. static
   C. const
   D. void

8. Which initialisation is valid for an integer array?
   A. int a = {1, 2, 3};
   B. int a[3] = (1, 2, 3);
   C. int a[3] = {1, 2, 3};
   D. array int a[3];

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9. Which declaration creates a floating-point variable named price?
   A. char price;
   B. int price;
   C. float price;
   D. bool price;

10. Which statement immediately exits a loop?
   A. break
   B. continue
   C. return 0
   D. cin

PART B (60 MARKS)

QUESTION 1

This question tests high-level languages and debugging.

a) Define syntax error and logic error.

(2 marks)

b) Explain TWO advantages of high-level languages over machine language.

(3 marks)

QUESTION 2

A concert charges RM15 for ages 0-12, RM25 for ages 13-17 and RM40 for ages 18 and above. A negative age is invalid.

a) Draw a flowchart for the ticket price selection.

(6 marks)

b) Write pseudocode and a complete C++ program for the ticket price selection.

(7 marks)

QUESTION 3

This question tests declarations, constants and expressions.

a) Write C++ declarations for: an integer age, a double price, a constant discount rate of 0.10 and a character grade.

(6 marks)

b) Write C++ statements to read age and price, calculate price after discount, and display the result.

(5 marks)

QUESTION 4

Study the following nested loop:

```
for (int i = 1; i <= 3; i++) {
   for (int j = 1; j <= i; j++) {
      cout << j;
   }
   cout << endl;
}
```

a) Trace the program and show the exact output.

(7 marks)

b) Identify TWO logic errors in a version that prints four rows but reuses j without resetting it. Rewrite the corrected code.

(8 marks)

QUESTION 5

A shop stores the prices of five items in an array.

a) Declare an array named prices with the values 20.5, 55.0, 15.9, 75.0 and 10.0. Display all values.

(8 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

(8 marks)

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PART C (30 MARKS)

QUESTION 1

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

QUESTION 2

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

END OF QUESTION PAPER